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Paleomagnetic results of Cretaceous cherts from Zhongba, southern Tibet: New constraints on the India-Asia collision

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ABSTRACT

An oceanic Greater Indian Basin was hypothesized to have formed between ~120 Ma and ~65 Ma by regional extension that separated the cratonic India from its northern margin of Tethyan Himalaya (TH). However, the lack of paleomagnetic data from this critical time interval in the TH marks an important gap in our knowledge about the tectonic evolution of the Greater India. We have conducted a paleomagnetic study of a Cretaceous marine succession in Zhongba, southern Tibet that consists of predominantly purple reddish cherts and siliceous limestone. Paleomagnetic samples from 12 sites were subjected to stepwise thermal demagnetization that revealed either two- or three-component magnetizations with the high-temperature component (HTC) typically unblocked at ~650 °C. The HTCs exhibit dual polarities that yield a positive reversed test at 95% confidence level, suggesting a primary origin of the remanence. With chronologic constraints from the combined analysis of the magnetostratigraphic and radiolarian data, the HTCs yield a mean of D_s = 336.9°, I_s = −50.7°, k_s = 18.0, α_{95} = 5.3° for the ~120.8–93.9 Ma interval of the TH, which corresponds to a paleopole at 25.0°N, 285.7°E (dp/dm = 4.8/7.1°). The apparent polar wander path (APWP) of the TH overall resembles that of cratonic India, suggesting no polarward separation that is sufficiently large to merit an oceanic basin between the TH and cratonic India. Therefore, our new reliable paleomagnetic data from the critical time interval do not support the Greater Indian Basin hypothesis and its implied dual collisional processes between India and Asia. On the other hand, the APWPs exhibit relative shifts for the post-120 Ma segment and the pre-130 Ma segment, indicating relative motions between the TH and cratonic India that most likely resulted from shearing along boundary-parallel faults during the India-Asia collision.

1. Introduction

The India-Asia collision represents one of the most significant geological events in the Cenozoic that raised the Tibetan Plateau and exerted profound impacts on the global atmospheric circulations (e.g., Molnar and Tapponnier, 1975; Dewey et al., 1988; Raymo and Ruddiman, 1992; Yin and Harrison, 2000; An et al., 2001; Guo, 2015). Widely accepted as an exemplified continent-continent collision, the India-Asia collision has long intrigued geoscientists with persisted debates regarding when, where, and how India and Asia collided (e.g., Besse et al., 1984; Patzelt et al., 1996; Yin and Harrison, 2000; Aitchison et al., 2007; Khan et al., 2009; Huang et al., 2010; Tan et al., 2010; Yi et al., 2011; Meng et al., 2012; van Hinsbergen et al., 2012; Bouilhol et al., 2013; Yang et al., 2015a,b; Hu et al., 2015, 2016; Huang et al., 2015a,b, 2017a,b,c; Ding et al., 2017; Najman et al., 2010; Aitchison and Ali, 2012; Sun et al., 2012; DeCelles et al., 2014). A traditional view depicts a single collision between India and Asia along the Indus-Tsangpo suture zone (ITSZ) in the late Paleocene or early Eocene (e.g., Molnar and Tapponnier, 1975; Patriat and Achache, 1984; Yin and Harrison, 2000). Subsequently, dual collisional processes were proposed that involved a collision of the India Plate with an intraoceanic arc at ~55 Ma prior to its final collision with Asia at ~35 Ma (Aitchison et al., 2007). More recently, a dual continent-continent collisional process was put forward in which the Greater India (Ali and Aitchison, 2005) is believed to consist of two continents, Tethyan Himalaya (TH) to the north and cratonic India to the south, separated by an oceanic Greater India Basin, and the TH first collided with Asia at ~50 Ma, followed by the closure of the oceanic Greater India Basin at ~25 Ma (van Hinsbergen et al., 2012).

The dual continent-continent collision model was proposed largely

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based on the paleomagnetic data of the TH and cratonic India. The pre-120 Ma paleomagnetic data of the TH and India yield similar paleolatitudes, whereas the ~70 to ~60 Ma paleomagnetic data of the TH show ~20° lower paleolatitudes than the expected coeval paleolatitude of India. As such, it was hypothesized that the TH must have rifted from cratonic India during the mid-Late Cretaceous, forming a ~ 2675 ± 700 km wide oceanic Greater India Basin (van Hinsbergen et al., 2012). Although high-quality Early Cretaceous paleomagnetic data of the TH have been reported and are used to support this model (Yang et al., 2015a; Ma et al., 2016), paleomagnetic data of the critical interval (~120–65 Ma), the mid-Late Cretaceous from the TH, that is essential for testing the model, remains lacking and represent a major gap in our knowledge. It has become even more urgent to obtain the mid-Late Cretaceous paleomagnetic data of the TH to test this model as the reliability of the ~70 to ~60 Ma paleomagnetic data of the TH, which was the basis for proposing the dual continent-continent collision model, is called into question (Huang et al., 2017a,b; Yi et al., 2017). The early Cenozoic paleopole of the TH was considered reliable previously and is now debated as to whether it is of primary or secondary origin (Huang et al., 2017a,b; Yi et al., 2017).

A recent paleomagnetic study of the Upper Cretaceous limestone of the Chuangde Formation from Gyangze area in eastern TH showed that the paleolatitude of the TH was similar to that of Indian craton during the interval from ~86.3 to 74.0 Ma (Zhang et al., 2017), suggesting the non-existence of a Greater India Basin during the Late Cretaceous and thus arguing against the "dual continent-continent" collision model. Because the Chuangde limestone was formed on a slope and likely underwent vertical axis rotations (Zhang et al., 2017), only inclinations are reliable and documented the paleolatitude of the TH. Clearly, more high-quality paleomagnetic data of the ~120–65 Ma interval from the TH are urgently needed to further test the dual continent-continent collision model. In addition, since the previous Late Cretaceous-Paleocene paleomagnetic data were derived mainly from eastern TH (Patzelt et al., 1996; Yi et al., 2011; Yang et al., 2015a; Ma et al., 2016; Zhang et al., 2017), i.e., the most proximal part of the TH, let alone previous paleomagnetic data that sufficiently meet the reliability criteria of Van der Voo (1990b) remain few, impeding our ability to fully understand its evolution. Therefore, the available paleomagnetic data from eastern TH cannot effectively constrain the northern extent of Greater India during the Cretaceous (Rowley and Ingalls, 2017). It is thus particularly desirable to obtain reliable paleomagnetic data of the critical time interval from other parts of the TH. Furthermore, given that carbonate rocks appear to be prone to remagnetization as demonstrated by numerous previous paleomagnetic studies of this region (Appel et al., 2012; Huang et al., 2015a, 2017a,b,c), we are keen on studying the paleomagnetic records of lithologies other than carbonate rocks to maximize the chance of obtaining reliable paleomagnetic data. Therefore, in this study, we have carried out a detailed paleomagnetic investigation of Cretaceous cherts in Zhongba area, which is situated in the central portion of the TH (Fig. 1a). The purpose of this study is to obtain high-quality paleomagnetic data from central TH and further test the dual continent-continent collision model.

2. Geological setting and sampling

The Indus-Tsangpo suture zone (ITSZ) marks the geological boundary between the northern margin of India and the southern margin of Asia (Fig. 1a). The TH is a tectonic unit situated at the northern part of the Greater India (Fig. 1a) and consists of two sub-units, separated by the Gyirong-Kangmar intracrustal thrust (Liu and Einsele, 1994; Zhang et al., 2012). The southern sub-unit is composed of shallow-water facies, while the northern sub-unit consists of deep-water facies. Zhongba area is located in the central part of the northern sub-unit (Fig. 1a) where the ITSZ splits into two branches confining the Zhongba micro-terrane (Li et al., 2014).

The study section (29.74°N, 83.99°E) is situated southwest to Zhongba county, near Route 219 (Fig. 1b). The outcrop consists of quartz sandstone, volcaniclastic sandstone and purple reddish cherts with limestone and siliceous limestone. These units were lumped together and assigned to the Cretaceous Sangdanlin Formation in the 1:250,000 geological map (Fig. 1b and c). Indeed, the lithological compositions and lithostratigraphic sequence of the outcrop are found to be correlative with those in Gyangze area in eastern TH. Specifically, the quartz sandstone (Unit 1, Fig. 1c), the volcaniclastic sandstone with spherical weathering (Unit 2, Fig. 1c), and the purple reddish cherts correspond to the Upper Jurassic Weimei Formation, the Lower Cretaceous Rilang Formation, and the (Upper) Gyabula Formation of Gyangze area, respectively. The age of the Weimei Formation was constrained to be Late Jurassic based on ammonite fossils (e.g., Wang et al., 1980; Liu, 1983). The Rilang Formation was considered to be late Early Cretaceous (Barremian-Aptian) based on the youngest detrital zircon U-Pb ages (e.g. Hu et al., 2008, 2010). The Gyabula Formation was well dated to the Aptian-Cenomanian with radiolarian, foraminifera, belemnite fossils (e.g., Li et al., 1999, 2005; Wang et al., 2000; Wan et al., 2005). The overlying dominantly purple reddish cherts interbedded or intercalated with marly limestone and shale are of Late Cretaceous.

We collected paleomagnetic samples from 12 sites from the purple reddish cherts dominated interval of the Zhongba succession. Each site contains 5 to 7 oriented hand samples taken from different stratigraphic beds and the spacing between the consecutive individual hand samples varies from 0.2 to 1.5 m. A total of 65 oriented hand samples were collected from the ~45 m thick interval in the Zhongba section (Fig. 1c). In the laboratory, the oriented hand samples were fixed with gypsum powder, drilled, and cut into the standard cylindrical specimens of 2.5 cm in diameter and 2.2 cm in height. Typically, one to three specimens were obtained from one oriented hand sample, i.e., from one bed. A total of 113 paleomagnetic specimens are obtained from the sampled interval of the Zhongba section.

3. Methods

Anisotropy of magnetic susceptibility (AMS) of all specimens was first measured with an AGICO KLY-3S Kappabridge. These paleomagnetic specimens were then subjected to stepwise thermal demagnetization up to 700 °C in an ASC-TD 48 thermal demagnetizer with an internal residual field less than 10 nT. Demagnetization temperature intervals are generally 100–50 °C in the low temperature range, and decrease to 30–10 °C in higher temperatures. Remanent magnetization was measured with a 2G-755 cryogenic magnetometer. All thermal demagnetization and remanence measurements were performed within a magnetically shielded room with a residual field less than 300 nT. Paleomagnetic data were analyzed using the computer program packages developed by Enkin (1990) and Cogné (2003). Demagnetization results were evaluated using stereographic projections and orthogonal diagrams (Zijderveld, 1967) and characteristic remanent magnetization (ChRM) directions of all the specimens were determined using the principal component analysis (Kirschvink, 1980), and the mean direction was calculated using Fisherian statistics (Fisher, 1953).

To examine magnetic properties of magnetic minerals, ten representative specimens from different beds were selected for rock magnetic experiments. Acquisition of isothermal remanent magnetization (IRM) was carried out by magnetizing the specimens with an ASC IM10-30 pulse magnetizer in a field ranging from 0.02 to 2.4 Tesla (T) and IRM intensity was subsequently measured following each step of magnetization with a JR-6A spinner magnetometer. Selected specimens were acquired composite IRM by applying a DC field of 2.4, 0.4 and 0.2 T sequentially along the Z-axis, Y-axis and X-axis of each specimen, respectively. Stepwise thermal demagnetization of the composite IRMs up to 700 °C (Lowrie, 1990) was performed in a SOGO-Fine TD-10 thermal demagnetizer and remanent magnetization was measured using a JR-6A spinner magnetometer. In addition, temperature-

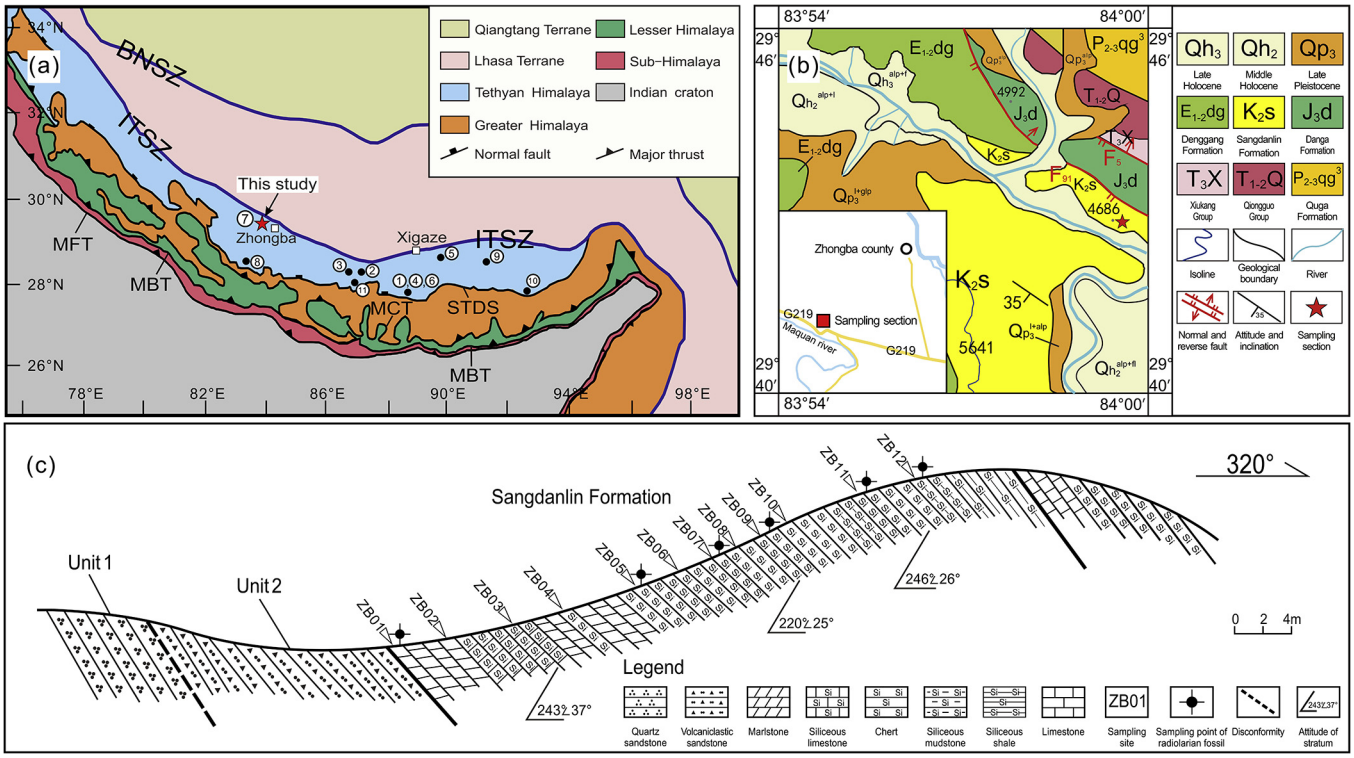


Fig. 1. Schematic map showing the geology and the sampling location of this study. (a) Regional geologic map of the Himalayan belt and adjacent areas (after Yang et al., 2015a); (b) Simplified geological map of the Zhongba area showing the stratigraphy and paleomagnetic sampling locations (after 1:250000 geological map of Babazhadong Sheet). (c) Cross section of the studied Zhongba section showing the lithology and the sampling scheme of this study. Abbreviations: BNSZ, Bangong-Nujiang suture zone; ITSZ, Indus-Tsangpo suture zone; STDS, South Tibet detachment system; MCT, Main Central thrust; MBT, Main Boundary thrust; MFT, Main Frontal thrust; The star indicates the Zhongha section of this study. The solid circles represent the sampling locations of previous paleomagnetic studies. ①, Patzelt et al. (1996) and Yi et al. (2011); ②, Besse et al. (1984); ③, Tong et al. (2008); ④, Patzelt et al. (1996); ⑤, Zhang BX et al. (2017); ⑥, Patzelt et al. (1996); ⑦, This study; ⑧, Klootwijk and Bingham (1980); ⑨, Ma et al. (2016); ⑩, Yang et al. (2015a); ⑪, Huang et al. (2015a).

dependent low-field magnetic susceptibility was measured from room temperature to 700 °C and then cool down to room temperature in an argon atmosphere using an AGICO KLY-3S Kappabridge with CS-3 high-temperature apparatus. To assess whether there is significant inclination shallowing in the study section, four oriented block samples whose orientation was measured on a bedding plane were selected and cut into 2 cm × 2 cm × 2 cm cubes to conduct a simplified magnetic anisotropy of isothermal remanence (AIR) experiment following Hodych and Buchan (1994)'s approach. The approach involves successively acquiring IRM up to 800 mT at 45° to the bedding and measuring the X-axis and Z-axis component of IRM at each IRM acquisition step, i.e., IRMx and IRMz. All paleomagnetic and rock magnetic experiments were performed at the Paleomagnetic Laboratory of Nanjing University.

To further constrain the age of the studied interval, six chert samples were collected for radiolarian micropaleontological analysis. In the laboratory, each sample was crushed into centimeter-size fragments and processed following the standard method (Dumitrica, 1970; Pessagno and Newport, 1972), treating with dilute hydrofluoric acid (HF). Relatively well-preserved radiolarian fossils were examined under a Hitachi TM3030 Scanning Electron Microscope (SEM) at the Key Laboratory of Economic Stratigraphy and Palaeogeography, Nanjing Institute of Geology and Palaeontology, Chinese Academy of Sciences.

4. Results

4.1. Anisotropy of magnetic susceptibility (AMS)

AMS data provide a measure of petrofabrics pertaining to deposition and/or stress. AMS results are graphically represented by an ellipsoid that is characterized by three principle axes (the maximum axis K_1, the intermediate axis K_2, the minimum axis K_3), the degree of anisotropy (P_j), and the shape factor T (prolate shape if $-1 < T < 0$; oblate shape if $0 < T < 1$) (Tarling and Hrouda, 1993). The AMS results of the study section show that the minimum axis (K_3) is approximately perpendicular to the bedding, and the intermediate axis (K_2) and maximum axis (K_1) are subparallel to the bedding (Fig. 2a). Both the F-L and T-P_j diagrams show that the study section is dominated by oblate fabrics with only a few being prolate fabrics (Fig. 2c and d), suggesting a dominant depositional-type fabric. The presence of few prolate fabrics and the slight tilt of K_3 off the vertical may indicate weak influence by current and/or stress.

4.2. Rock magnetic results

Various rock magnetic data are used to identify magnetic minerals in the samples based on their characteristic coercivities, unblocking temperatures, and other diagnostic thermomagnetic properties (Dunlop and Özmer, 1997). The IRM acquisition curves show that the samples do not reach saturation at ~800 mT or above (Fig. 3a), indicative of the dominance of high-coercivity magnetic minerals in the samples. Demagnetization of the composite IRMs of marlstone from the lower part of the section shows that the hard, intermediate, and soft components were completely unblocked at ~580 °C (Fig. 3b), indicating that magnetite is the major remanence carrier of these samples. For cherts and siliceous limestone, thermal demagnetization of the composite IRMs shows a gradual decrease in the intensity of all the three components between 550 °C and 600 °C, followed by a sharp decrease at ~680 °C (Fig. 3c and d), suggesting that remanence resides in both magnetite and hematite. The abundant occurrence of hematite in the samples is also revealed by the thermomagnetic properties of the samples that

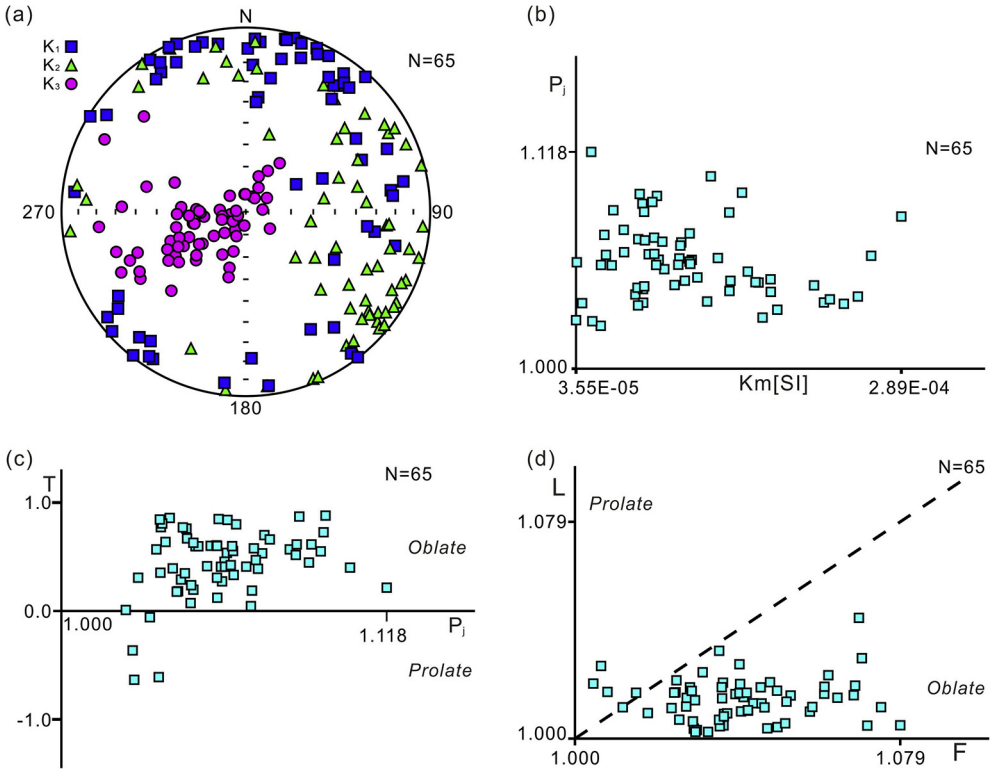


Fig. 2. Anisotropy of magnetic susceptibility (AMS) results. (a) Equal-area projections of the AMS data in stratigraphic coordinates; (b) Plot of the degree of anisotropy (P_j) versus the mean magnetic susceptibility (K_m); (c) Plot of the shape factor (T) versus the degree of anisotropy (P_j); (d) the F-L diagram of the specimens, L, lineation = K_1/K_2; F, foliation = K_2/K_3; N, number of specimens measured.

exhibit a distinct drop of magnetic susceptibility between 600 °C and 680 °C during heating (Fig. 4). The pronounced increase of magnetic susceptibility between ~600 °C and ~500 °C during cooling (Fig. 4) indicates the occurrence of mineralogic transformation, likely producing magnetite, during heating.

The anisotropy of isothermal remanence (AIR) results are shown in Fig. 5. Since magnetite has been largely saturated by 200 mT, the ratio IRMz/IRMx of IRMs acquired between 200 and 800 mT was calculated to estimate the IRM anisotropy contributed by hematite. The magnetic anisotropy of hematite is determined using the slope of the best-fit line for data points between 200 mT and 800 mT (Fig. 5). According to tan I_o/tan I_f = IRMz/IRMx, where I_o is the observed inclination of

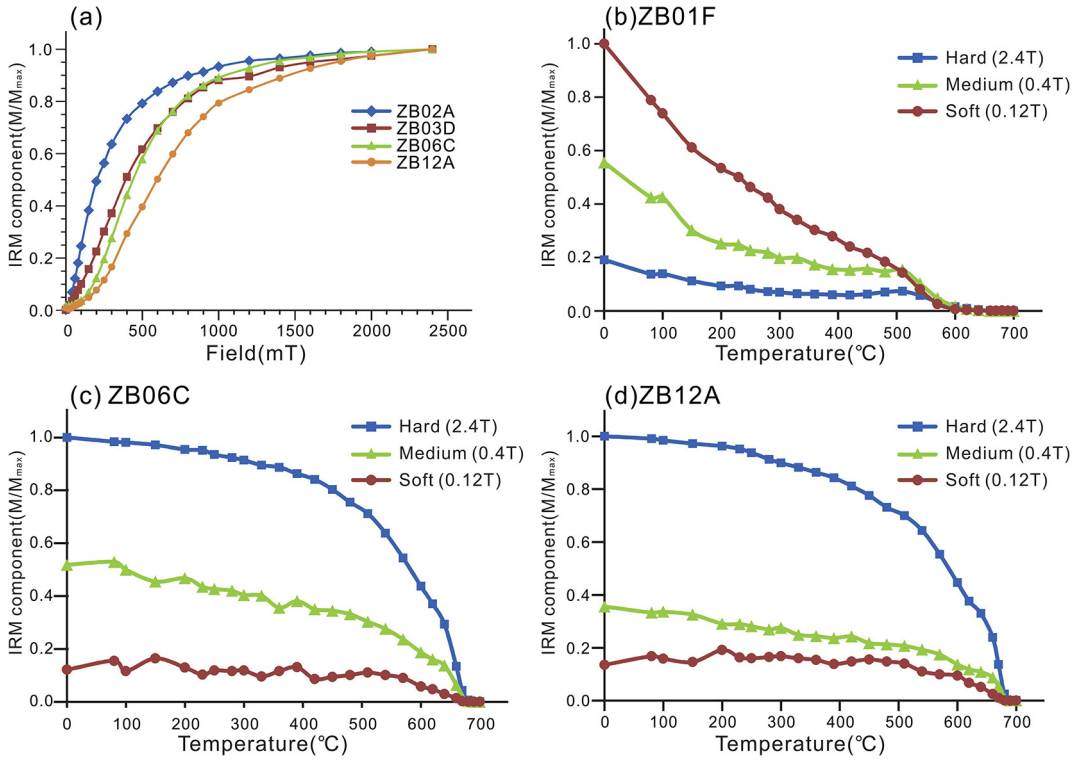


Fig. 3. Isothermal remanent magnetization (IRM) acquisition curves (a) and thermal demagnetization results of the composite IRM acquired sequentially along three orthogonal axes (b-d) from the study section.

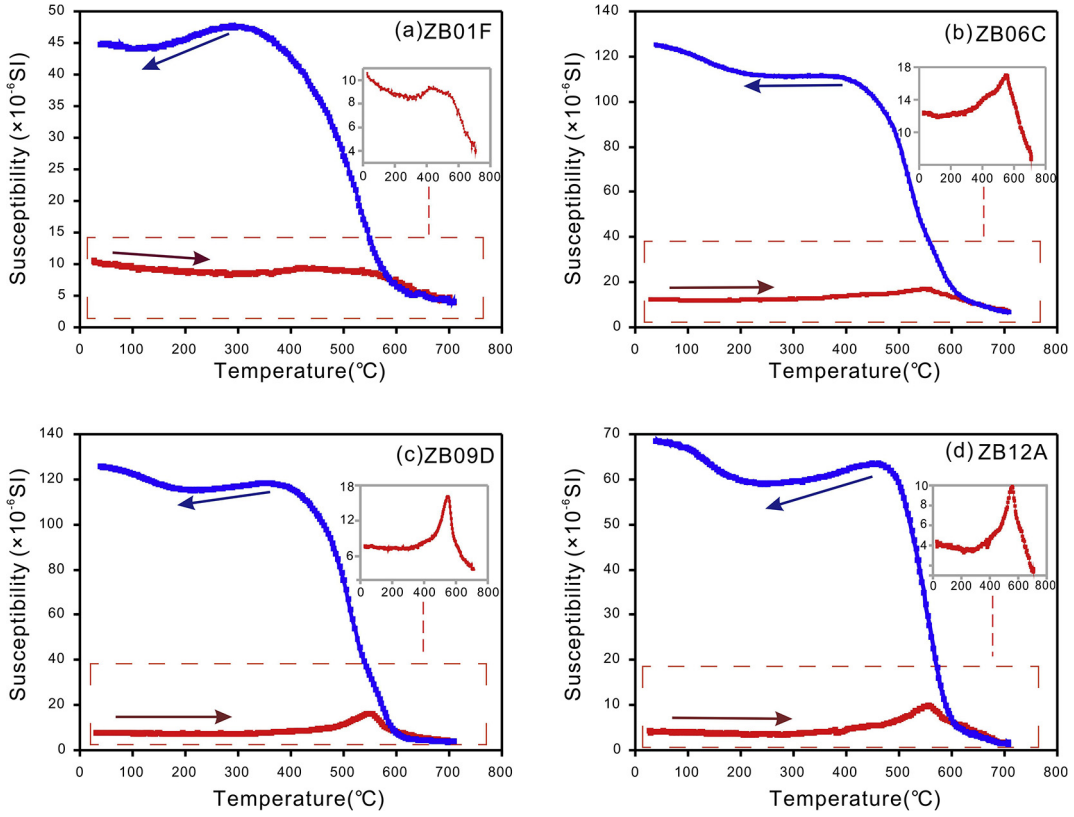


Fig. 4. Temperature-dependent low-field magnetic susceptibility curves of the representative specimens from the study section.

remanence and I_f is the inclination of the ambient field, these specimens experienced only 1.1–1.7° of inclination shallowing (Table 1), suggesting that there was not significant inclination shallowing in our study section.

4.3. Radiolarian results

Poorly to moderately preserved radiolarians were extracted from 6 cherty samples, and only 9 indeterminable species belonging to 9 genera can be identified (Table 2, Fig. 6). Among six samples collected for radiolarian fossil analysis, two samples, 17ZB05A (20.0 m) and 17ZB12B (43.5 m), contain radiolarian fossils that cannot be readily and positively identified. The other four samples contain radiolarian fossils that are identifiable and were used to constrain the ages. Below is the detailed documentation of the radiolarian fossil data from the four samples.

Sample 17ZB01A was collected from 0.0 m and contains *Archaeodictyomitra* (?) sp., *Hiscocapsa* sp., *Pseudodictyomitra* (?) sp., *Pseudoeucyrtis* (?) sp. and *Svinitzium* (?) sp. (Table 2, Fig. 6). According to O'Dogherty et al. (2009), *Archaeodictyomitra* ranged from early Pliensbachian of Early Jurassic to late Campanian of Late Cretaceous, *Hiscocapsa* ranged from late Bathonian of Middle Jurassic to middle Albian of Early Cretaceous, *Pseudodictyomitra* ranged from early Kimmeridgian of Late Jurassic to Tertiary, *Pseudoeucyrtis* ranged from Late Triassic to early Turonian of Late Cretaceous, and *Svinitzium* only occurred in Early Cretaceous, spanning early Berriasian to late Albian. The assemblage indicates that the age of this sample is early Berriasian to middle Albian.

Only two indeterminable species, *Becus* (?) sp. and *Godia* sp., can be recognized in Sample 17ZB07D that was collected at 27.9 m. *Becus* occurred in the interval between late Kimmeridgian of Late Jurassic and early Turonian of Late Cretaceous (O'Dogherty et al., 2009), while *Godia* ranged from early Berriasian of Early Cretaceous to late Cenomanian of Late Cretaceous (O'Dogherty et al., 2009). So the co-occurrence of these two species indicates early Berriasian of Early Cretaceous–late Cenomanian of Late Cretaceous in age.

Sample 17ZB09D was collected from 31.8 m and contains *Becus* (?) sp., *Cavidiscus* (?) sp. and *Godia* sp.. The *Cavidiscus* (?) specimens are tentatively assigned into genus of *Cavidiscus*.. The age of this sample is early Berriasian of Early Cretaceous–late Cenomanian of Late Cretaceous according to O'Dogherty et al. (2009).

Godia sp. and *Xitus* sp. can be identified in Sample 17ZB11B at 39.1 m. According to O'Dogherty et al. (2009), *Xitus* occurred in late Bajocian of Middle Jurassic to late Campanian of Late Cretaceous. The assemblage suggests that the age of this sample is early Berriasian of Early Cretaceous–late Cenomanian of Late Cretaceous.

Collectively, the age of the radiolarian fossil assemblages from the investigated strata ranges from Berriasian (Early Cretaceous) to Cenomanian (Late Cretaceous) (~145–93.9 Ma) (Ogg and Hinnov, 2012).

4.4. Paleomagnetic results

About 15% specimens yield erratic directions upon thermal demagnetization (Fig. 7a). About 25% specimens show stable demagnetization trajectories at low and intermediate temperature ranges, but exhibit erratic directions at high temperatures and thus characteristic remanent magnetization cannot be isolated (Fig. 7b and c). The remaining ~60% specimens display either dual-component (Fig. 7d and e) or three-component remanent magnetizations (Fig. 7f–i). For the dual-component remanent magnetizations, component 1 (C1) can be cleaned by ~300–580 °C, and component 2 (C2) can be isolated from above those temperatures up to ~680 °C (Fig. 7d and e). Component C2 corresponds to the high temperature component (HTC) and decays toward the origin, defining characteristic remanent magnetizations (ChRMs). For the three-component remanent magnetizations, the low temperature component (LTC) was typically removed by ~250–380 °C and the intermediate temperature component (ITC) was separated by

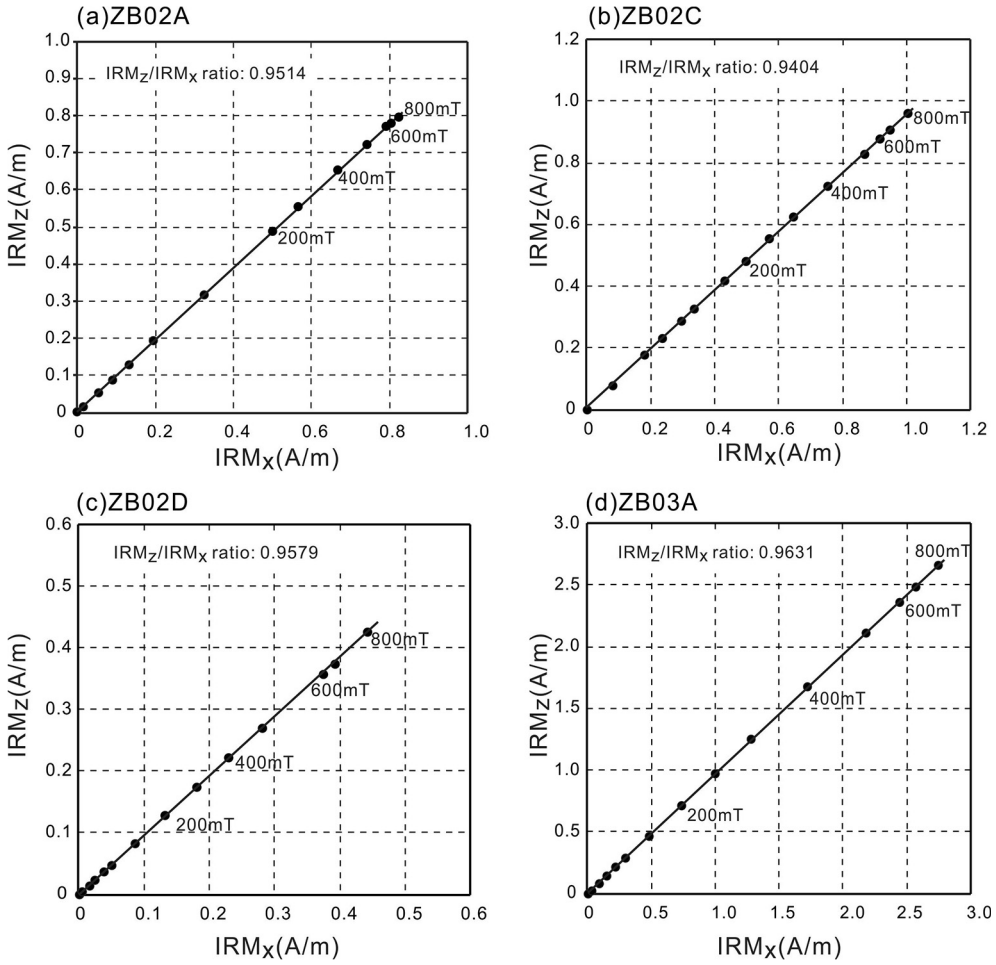


Fig. 5. Plot of IRMz versus IRMx showing the gradient of the best-fitting lines to estimate the anisotropy of isothermal remanence (AIR) of hematite.

Table 1
The anisotropy of isothermal remanence (AIR) results of the samples from the study section

Specimen ID	Inclination (°)	IRMz/IRMx (acquired between 200 and 800 mT)	Inclination corrected (°)	Inclination shallowing (°)
ZB02A	−55.3	0.9514	−56.6	1.3
ZB02C	−51.2	0.9404	−52.9	1.7
ZB02D	−41.8	0.9579	−43.0	1.2
ZB03A	45.6	0.9631	46.7	1.1

~400 °C and ~600 °C. The demagnetization trajectories at high temperatures decay linearly toward the origin and define the HTCs, or ChRMs, typically between ~600 °C and ~680 °C (Fig. 7f–i). The three-component remanent magnetizations mainly occur in specimens from Sites ZB04 to ZB11 that are predominantly composed of reddish cherts.

The mean direction of the LTCs in the geographic coordinates is $D_g = 355.7^\circ$, $I_g = 51.3^\circ$, $k_g = 10.6$, $\alpha_{95} = 5.5^\circ$, which is close to the present-day geomagnetic field of the study area (Dec = −0.5°, Inc = 48.7°) (Fig. 8a), suggesting that the LTCs likely represent the overprint of the present-day geomagnetic field. The HTCs, or the ChRMs, show both northerly upward and southerly downward directions (Fig. 8c and d). The mean of the northerly upward directions is $D_{1s} = 339.7^\circ$, $I_{1s} = -51.5^\circ$, n = 49, k = 18.3, and the mean of the southerly downward directions is $D_{2s} = 186.1^\circ$, $I_{2s} = 51.3^\circ$, n = 4, k = 61.2. The means of the northerly upward and southerly downward directions were used to perform a reversed test and yielded $r_0 = 16.4^\circ < r_c = 17.1^\circ$, indicating a positive reversal test at 95% confidence level in classification "C" (McFadden and McElhinny, 1990). In addition, the northerly upward and southerly downward directions become more near anti-parallel in stratigraphic coordinates than in

Table 2
The radiolarian fossils identified from the investigated strata.

Sample ID	Height (m)	Genera	Age
17ZB01A	0.0	*Archaeodictyomitra* (?) sp., *Hiscocapsa* sp., *Pseudodictyomitra* (?) sp., *Pseudoeucyrtis* (?) sp., *Svinitzium* (?) sp.	Early Berriasian–middle Albian
17ZB07D	27.9	*Becus* (?) sp. and *Godia* sp.	Early Berriasian–late Cenomanian
17ZB09D	31.8	*Becus* (?) sp., *Cavidiscus* (?)., *Godia* sp.	Early Berriasian–late Cenomanian
17ZB11B	39.1	*Godia* sp. and *Xitus* sp.	Early Berriasian–late Cenomanian

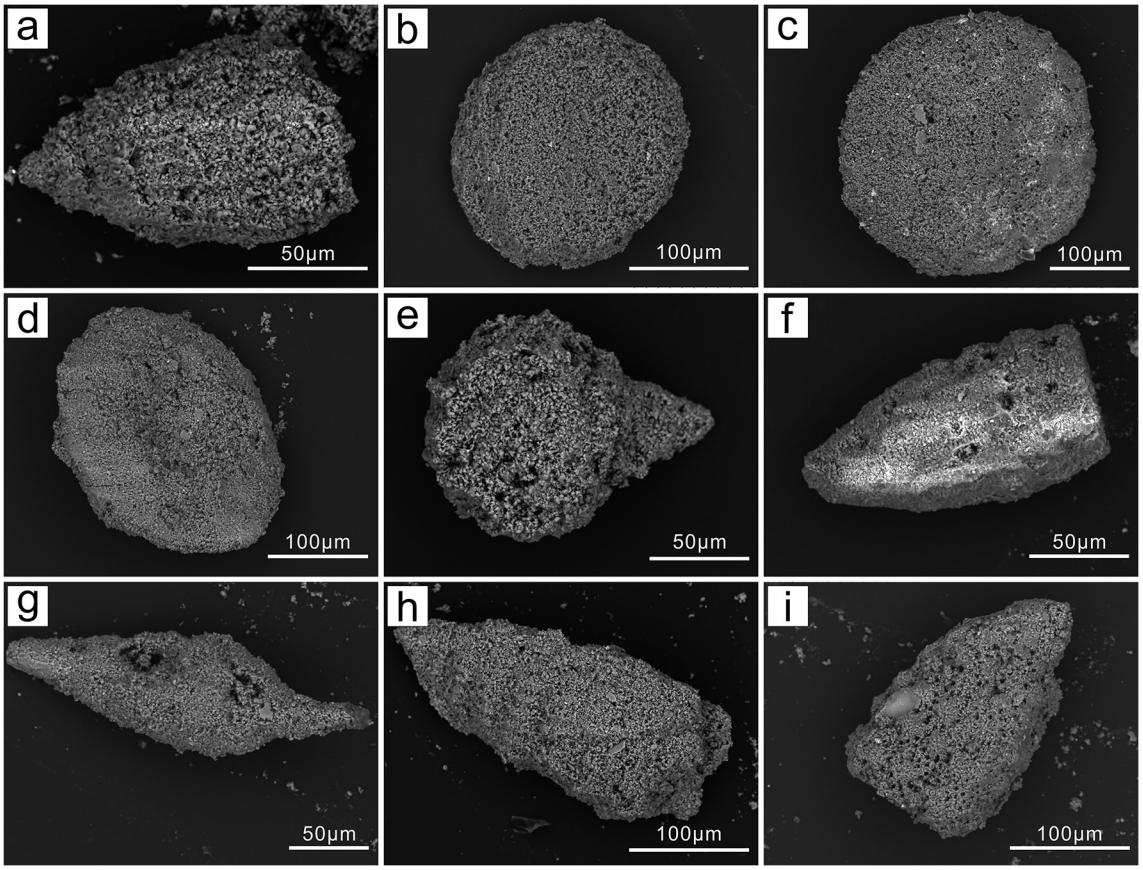


Fig. 6. SEM photomicrographs of the radiolarian fossils from the Zhongba cherty strata. (a) *Archaeodictyomitra* (?) sp.; (b) *Becus* (?) sp.; (c) *Cavidiscus* (?) sp.; (d) *Godia* sp.; (e) *Hiscocapsa* sp.; (f) *Pseudodictyomitra* (?) sp.; (g) *Pseudoeucyrtis* (?) sp.; (h) *Svinitzium* (?) sp.; (i) *Xitus* sp.

geographic coordinates, suggesting that the remanence was acquired prior to tilting. Since tilting was most likely associated with the India-Asia collision in the early Cenozoic, the improved anti-parallelism of two groups of directions after tilt-correction suggests a positive tilt test. Thus the positive reversal test and tilt test suggest that the magnetizations were likely primary. The mean in geographic coordinates is $D_g = 338.3^\circ$, $I_g = -28.5^\circ$, $k_g = 14.5$, $\alpha_{95} = 5.3^\circ$ (Fig. 8c) and becomes $D_s = 341.7^\circ$, $I_s = -51.7^\circ$, $k_s = 18.4$, $\alpha_{95} = 4.7^\circ$ after tilt correction (Fig. 8d), corresponding to a paleopole at 25.5°N, 281.1°E (dp/dm = 4.4°/6.4°).

5. Discussions

5.1. Magnetostratigraphy of the Zhongba section

The ChRMs of the study section exhibit both northerly upward directions and southerly downward directions that passed a reversal test at 95% confidence level in classification "C", suggesting that the remanent magnetizations were likely primary. Radiolarian fossil data show that the study section spans Berriasian to Cenomanian (Table 2), an interval shortly after the initial break-off of India Plate from Gondwana in Early Cretaceous (Zhu et al., 2009; Olierook et al., 2016). Therefore, the study section was accumulated when the India Plate was drifting from high to low latitudes in the Southern Hemisphere. As such, the northerly upward directions recorded the normal polarity intervals of the geomagnetic field and the southerly downward directions documented the reversed polarity intervals of the geomagnetic field. The ChRMs of different stratigraphic height are summarized in Table 3 and are graphically shown in Fig. 9. One striking feature is that the ChRM data show persistent normal polarities in the upper segment (> ~10 m Height) of the study section, indicating a protracted long normal polarity interval (Fig. 9). Two beds in the lower segment (< ~10 m Height) of the study section display reversed polarities (Fig. 9). However, the paucity of the reversed polarity data and the fact that these reversed polarities do not occur consecutively do not allow to define reversed polarity zones with confidence. In essence, the study section is predominated by normal polarities.

Inclination shallowing is one of major concerns for the accuracy of ChRMs of sedimentary rocks. Unlike siliciclastic rocks that are prone to compaction-induced inclination shallowing (Kodama, 2012), cherts are of biological and/or chemical origin, and are less susceptible to compaction-induced inclination shallowing. Previous paleomagnetic studies of Triassic and Jurassic cherts from central Japan show that inclinations of cherts were not much flattened by compaction (Shibuya and Sasajima, 1986; Ando et al., 2001), 2° at most (Oda and Suzuki, 2000). For the red cherts in our study section, the preserved radiolarian fossils show no signs of deformation (Fig. 6), suggesting no significant compaction-induced inclination shallowing. The simplified anisotropy-based inclination correction shows that inclination shallowing of the studied red cherts is only 1.1° to 1.7° (Table 1), which is well within the 95% confidence interval. Therefore, the effect of inclination shallowing on the ChRMs of the study section is minimal and negligible.

The chronology of the study section can be constrained by the combined paleomagnetic, radiolarian, and lithostratigraphic data. The radiolarian fossil assemblages indicate a broad age range spanning Berriasian (Early Cretaceous) to Cenomanian (Late Cretaceous) (Fig. 9). The volcanolcastic sandstones underlying the study section represent the Cretaceous Rilang Formation that is constrained at ~127.7 Ma (Hu et al., 2008). Since the paleomagetic data show a predominance of normal polarities, the chronology of the study section is either entirely or partially within the Cretaceous normal polarity superchron C34n (Ogg and Hinnov, 2012) (Fig. 9). Since the base of M0r was dated at

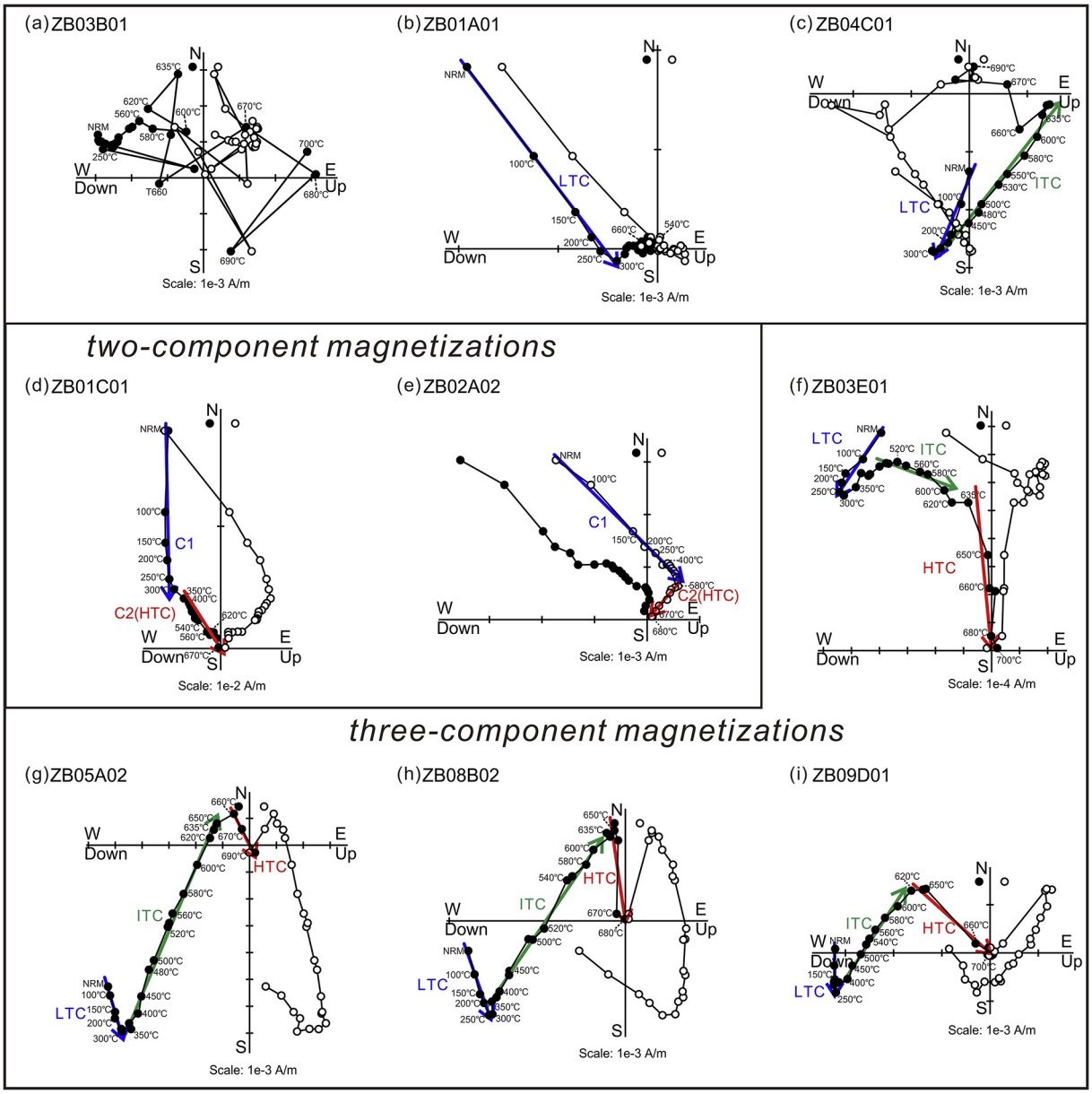


Fig. 7. Thermal demagnetization diagrams of the representative specimens in geographic coordinates. The solid and open symbols represent the projections onto the horizontal and vertical planes, respectively. The blue, green, and red arrows denote the LTC, ITC, and HTC, respectively. Note that "Up" points to the right in order to best illustrate the demagnetization trajectories. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

121.2 ± 0.5 Ma (He et al., 2008) and M0r lasted for about 0.4 Myr (Ogg and Hinnov, 2012), the base of C34n is estimated to be ~120.8 Ma (Fig. 9). The age of the top of C34n is known to be 83.6 Ma (Ogg and Hinnov, 2012). Because the polarity zones cannot be defined with confidence for the lower segment (< ~10 m Height) of the study section, we focus on the upper segment of the study section that displays a prolonged normal polarity interval as highlighted in a gray band, or N1 zone, in Fig. 9. With the above-mentioned combined constraints, the upper segment, i.e., N1 zone, of the study section must have been within the superchron C34n.

The chronology of the N1 zone can be further refined. Radiolarian fossil data suggest that the top of the N1 zone reaches Cenomanian that spans 100.5 to 93.9 Ma. 93.9 Ma is taken as a conservative estimate for the upper bound age for the N1 zone, although the top of the N1 zone could be older than 93.9 Ma. For the base of the N1 zone, because the polarity zones cannot be defined with confidence for the lower segment underlying the N1 zone, the maximum age of the base of the N1 zone could be as old as the basal age of the superchron C34n, i.e., 120.8 Ma. Therefore, 120.8 Ma is taken as a conservative estimate for the base of the N1 zone, although it could be younger than 120.8 Ma. Taken together, the chronology of the N1 zone is estimated to range from ~120.8 to 93.9 Ma with a mean of ~ 110.0 Ma. ChRMs of the N1 zone yield a mean of D_s = 336.9°, I_s = -50.7°, k_s = 18.0, α_{95} = 5.3° after tilt correction, corresponding to a paleopole at 25.0°N, 285.7°E (dp/dm = 4.8/7.1°) at ~110 Ma for Tethyan Himalaya.

5.2. Implications for the India-Asia collision

The oceanic Greater India Basin hypothesis was put forward based on the comparison of paleolatitudes of the TH and the cratonic India (van Hinsbergen et al., 2012). The similar paleolatitudes at ~120 Ma and the ~20° latitudinal difference at ~65 Ma, led van Hinsbergen et al. (2012) to hypothesize a N-S extension in Cretaceous from ~120 to 65 Ma, forming an ~2675 ± 700 km oceanic Greater India Basin.

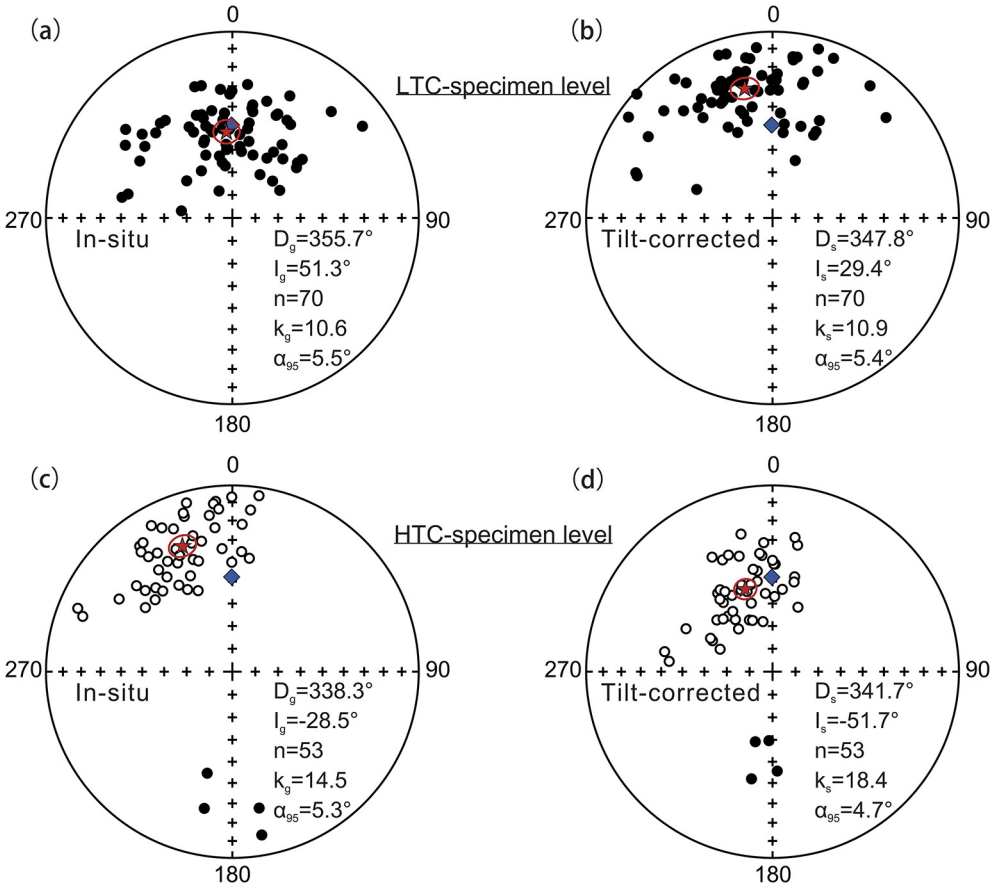


Fig. 8. Equal-area projections of the low-temperature components (LTCs) and high-temperature components (HTCs) of the representative specimens from the study section before (a, c) and after (b, d) tilt correction. The solid/open circles represent downward/upward inclinations; the red stars indicate the overall mean directions; the blue rhombuses indicate the present-day geomagnetic field of the study area. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

Table 3
Paleomagnetic results of the study section.

Height (m)	Lithology	Specimen ID	Strike(°)/Dip(°)	Dg (°)	Ig (°)	Ds (°)	Is (°)	α_{95} (°)	VGP-latitude (°)
44.0	Siliceous shale	ZB12D	272/23	355.2	−13.4	353.8	−36.3	3.2	39.7
43.1	Siliceous shale	ZB12A	272/23	8.6	−5.0	9.5	−27.8	7.6	44.6
38.9	Chert	ZB11A	246/26	324.1	−33.0	316.9	−58.2	6.6	10.5
37.6	Chert	ZB10E	230/18	335.7	−32.1	340.5	−49.2	2.9	27.4
34.7	Chert	ZB10C	230/18	331.4	−48.4	338.7	−65.9	6.2	9.8
33.2	Chert	ZB10B	230/18	337.1	−51.0	349.2	−67.7	6.7	9.1
31.8	Siliceous mudstone	ZB09D	220/25	−24.6	329.3	329.3	−48.6	3.0	23.9
31.6	Chert	ZB09B	220/25	355.2	−34.1	13.5	−49.0	5.2	29.0
31.1	Chert	ZB09A	220/25	343.0	−37.9	1.6	−56.8	3.1	22.9
30.6	Chert	ZB08E	247/22	340.8	−36.4	342.8	−58.4	10.0	19.3
29.3	Chert	ZB08C	247/22	342.8	−13.8	343.9	−35.7	3.6	38.1
28.8	Chert	ZB08B	247/22	351.6	−16.3	354.8	−37.4	4.1	39.1
27.9	Chert	ZB07D	246/21	6.4	−13.7	11.2	−31.4	3.1	42.1
26.6	Chert	ZB06E2	233/20	331.0	−11.3	332.2	−31.1	10.4	36.3
26.3	Chert	ZB06E1	233/20	343.8	−25.0	349.3	−43.4	8.8	34.0
25.8	Chert	ZB06D	233/20	329.6	−31.9	332.0	−51.7	10.7	22.5
25.2	Chert	ZB06C	233/20	344.8	−6.8	347.1	−25.2	8.8	45.3
24.6	Chert	ZB06B1	233/20	306.4	−41.6	297.3	−60.3	7.2	−1.6
21.6	Chert	ZB05C	220/20	317.6	−38.9	321.4	−58.7	3.7	12.1
20.0	Chert	ZB05A	220/20	333.3	−35.0	342.4	−52.8	6.7	24.7
17.9	Marlstone	ZB04D	235/37	290.3	−15.5	275.7	−43.6	6.7	−7.8
12.7	Marlstone	ZB03E	246/35	354.4	−9.0	0.7	−41.7	4.9	36.2
11.2	Marlstone	ZB03D	230/30	323.9	−18.2	325.5	−48.1	3.4	22.6
9.2	Siliceous limestone	ZB03A	242/36	169.8	11.9	177.3	45.6	12.6	−33.2
9.0	Siliceous limestone	ZB02E	250/28	3.8	−21.8	13.1	−46.6	2.7	31.0
7.0	Siliceous limestone	ZB02D	265/35	359.8	−6.9	1.4	−41.8	11.7	36.2
5.5	Siliceous limestone	ZB02C	245/33	337.6	−18.2	338.9	−51.2	3.4	25.2
4.5	Marlstone	ZB02B	284/15	193.6	43.3	193.4	58.3	4.0	−20.1
4.0	Marlstone	ZB02A	284/15	359.7	−40.9	354.8	−55.3	3.9	24.2
0.8	Marlstone	ZB01C	244/27	323.2	−40.7	313.0	−66.8	2.8	0.5

Notes: Specimen ID, Specimen identification; Strike/Dip, strike of bed, following the right-hand rule; Dip, dip of bed; Dg, Ig, Ds, Is, declination and inclination in geographic and stratigraphic coordinates, respectively; α_{95}, 95% confidence limit of Fisher statistics after tilt correction; VGP-latitude, latitude of corresponding virtual geomagnetic pole.

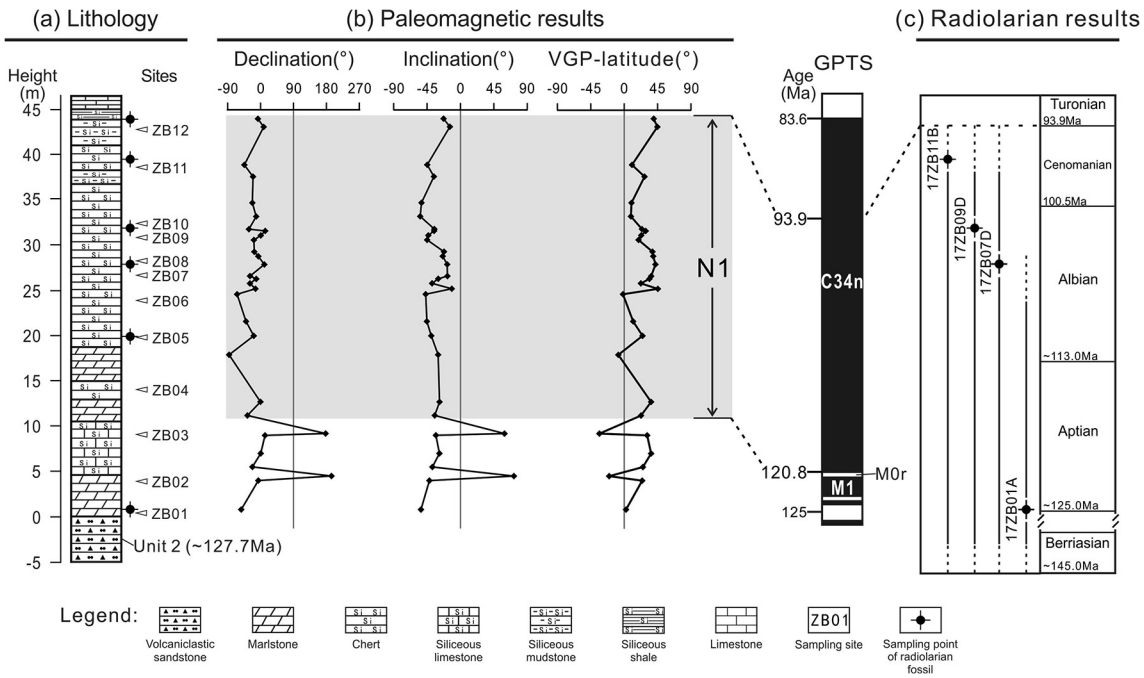


Fig. 9. A summary of the lithostratigraphic, magnetostratigraphic, and radiolarian results from the Zhongba section that collectively bracket the prolonged normal polarity zone, i.e., N1, to the ~120.8 Ma to 93.9 Ma interval. Unit 2 (Figs. 1c and 9a) belongs to the Rilang Formation that was constrained to ~127.7 Ma (Hu et al., 2008); GPTS, geomagnetic polarity time scale (Ogg and Hinnov, 2012).

The hypothesis also entails that the TH collided with Asia at ~50 Ma and then the oceanic Greater India Basin closed at ~25 Ma, which marked the completion of the India-Asia collision. It is clear that the hypothesis hinges on the accuracy of the estimate for the N-S extension of the Greater India. The northern extent of the Greater India was subsequently disputed because India Plate underwent a nearly 90° counterclockwise rotation as it drifted northwards prior to its collision with Asia (Wang et al., 2014), which could lead to inaccurate estimate of the northern extent of the Greater India. Huang et al. (2015b) examined the "Greater India Basin" extension by taking into account the Triassic to Cretaceous reconstruction of India and concluded that the counterclockwise rotation cannot explain the latitudinal difference between the TH and the cratonic India. However, as pointed out by Rowley and Ingalls (2017), the pre-120 Ma reconstruction cannot constrain the hypothesized extension during the mid-Late Cretaceous as the ~90° counterclockwise rotation of India Plate occurred mainly after 120 Ma. Zhang and Huang (2017) investigated the impact of the ~90° counterclockwise rotation on the estimates of latitudinal difference between the TH and the cratonic India during the Cretaceous. They suggested that the varied paleolatitudinal differences between the TH and cratonic India were a result of the nearly 90° counterclockwise rotation if the India Plate is considered a rigid or semi-rigid block; while the Great India Basin scenario is permissible, but not unique, if the TH and the cratonic India are considered as two individual blocks.

To contribute to an improved understanding of the tectonic evolution of Greater India, this study produced new paleomagnetic data of ~120.8 to 93.9 Ma rocks from central part of the TH. The new paleomagnetic data fill in the gap in the paleomagnetic dataset of ~120–65 Ma of the TH, a critical time interval when an oceanic Greater India Basin was hypothesized to have formed (van Hinsbergen et al., 2012), and thus shall provide important constraints on the Greater India Basin hypothesis. As an oceanic basin is hypothesized between the TH and the cratonic India, the TH and the cratonic India is implicitly viewed as two individual continental blocks, which in paleomagnetism are commonly considered as rigid blocks. Therefore, the Greater India Basin hypothesis can be tested by comparing the apparent polar wander paths (APWP) of the two blocks just as the classic example of the opening of the Atlantic Ocean dictated by the separate, but similar patterns of the APWPs of North America and Eurasia (Van der Voo, 1990a). The APWP of the cratonic India is derived from Torsvik et al. (2012) as has also been used by van Hinsbergen et al. (2012). To construct the APWP of the TH, besides our new paleomagnetic data of ~120.8–93.9 Ma from this study, we compiled the available paleopoles from the TH whose ages straddle the key time interval of ~120 Ma to 65 Ma in order to effectively test the Great India Basin hypothesis (van Hinsbergen et al., 2012) (Table S3). The reliability of these paleopoles was assessed with Van der Voo (1990b)'s criteria. Paleopoles with the reliability factor Q ≥ 5 are considered acceptable for constructing the APWP of the TH. In addition, for constructing an APWP, the criterion "tectonic coherence with craton or block involved" is crucial. Among the 9 paleopoles with Q ≥ 5, the paleopole from the Chuangde Fm was affected by local vertical-axis rotation (Zhang et al., 2017) and is thus excluded. Also, because two Paleocene paleopoles with the reliability factor Q = 5 from Dingri area were obtained from paleomagnetic datasets that are conisdered to be too small by modern standards, thse two paleopoles are excluded. Although the paleopole from Dzong (Q = 5) lacks a field stability test (Klootwijk and Bingham, 1980), it is included mainly because its age is within this key time interval and shows "tectonic coherence" with the TH. But this pole is highlighted with a slightly light color in the APWP of the TH (Fig. 10). So, the newly constructed APWP of the TH consists of 6 paleopoles and is shown in Fig. 10.

The most salient feature is that the pattern of the TH's APWP is similar to that of the cratonic India's and the two APWPs are largely overlapped polarward. In particular, the new paleomagnetic data from the TH overlap the APWP of cratonic India. These results suggest no significant separation between the TH and cratonic India since ~120 Ma. In other words, the data do not show paleomagnetically detectable separation between the TH and the cratonic India that merits a ~ 2675 +/−700 km wide oceanic basin. Together with numerous lines of geological evidence (Hu et al., 2016 and reference thereafter), the new paleomagnetic data do not support a Greater India Basin hypothesis. Therefore, the new paleomagnetic data from the critical time interval of ~120–65 Ma do not support the dual collision processes

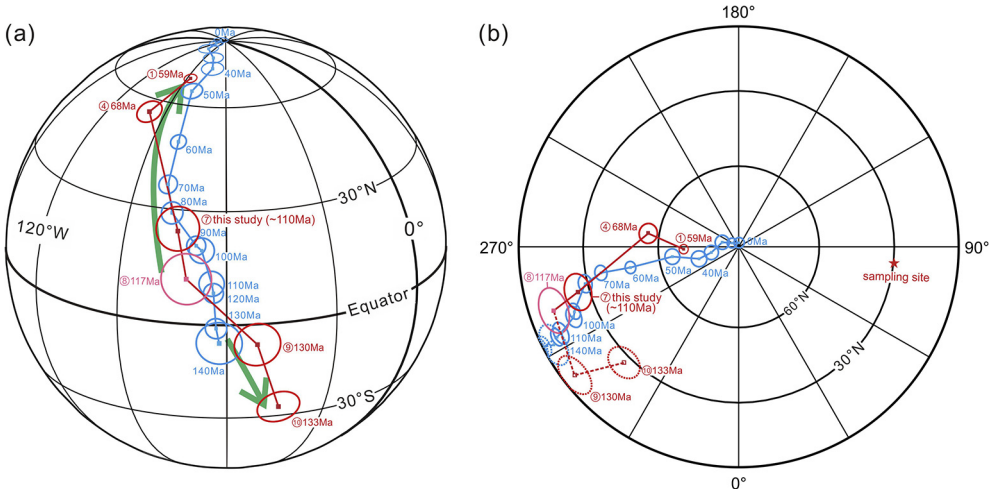


Fig. 10. Comparison of the apparent polar wander paths (APWPs) of Tethyan Himalaya (TH, in red) and cratonic India (in blue, Torsvik et al., 2012). The numbers in circles indicate the sources of paleomagnetic data from Tethyan Himalaya: ①, Patzelt et al. (1996) and Yi et al. (2011); ④, Patzelt et al. (1996); ⑦, This study; ⑧, Klootwijk and Bingham (1980); ⑨, Ma et al. (2016); ⑩, Yang et al. (2015a). (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

between India and Asia as envisioned by the Greater India Basin hypothesis, but favor the traditional single collision model.

Although the general patterns of the two APWPs are largely similar, coeval paleopoles do not coincide, and mostly show a relative shift. The ~120 Ma and younger paleopoles of the TH show a northward shift relative to the coeval paleopoles of the cratonic India, while the 130 Ma and older paleopoles of the TH display a southward shift relative to the coeval paleopoles of the cratonic India (Fig. 10). The relative shifts between the APWPs of the two blocks (green arrows in Fig. 10) suggest relative longitudinal motions between the TH and the cratonic India. The relative motions between the TH and the cratonic India can be readily accommodated by sinistral or dextral shearing after the initial India-Asia collision along faults generally parallel to the boundary of the TH and the cratonic India.

6. Conclusions

We carried out a detailed paleomagnetic study of Cretaceous cherts in Zhongba, southern Tibet. The stable ChRMs were obtained by means of stepwise thermal demagnetization and the ChRMs yielded a positive reversal test at the 95% confidence level, suggesting that the ChRMs are primary in origin. The magnetostratigraphy and radiolarian data constrain the age of the main part of the sampled strata at ~120.8–93.9 Ma. The mean of the ChRMs is $D_s = 336.9°$, $I_s = -50.7°$, $k_s = 18.0$, $\alpha_{95} = 5.3°$ after tilt correction, corresponding to a paleopole at 25.0°N, 285.7°E (dp/dm = 4.8°/7.1°). The similar patterns of the APWPs of the TH and cratonic India suggest that these two blocks were not significantly separated and it was thus unlikely separated by a ~ 2675 +/− 700 km wide oceanic basin between the TH and cratonic India during the mid-Late Cretaceous as speculated in the Greater India Basin hypothesis. The relative shifts of the portions of the APWPs may indicate relative motions between the TH and cratonic India during the India-Asia collision.

Declaration of interest

None declared.

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Appendix A. Supplementary material

Supplementary data to this article can be found online at https://doi.org/10.1016/j.jseaes.2019.01.012.

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